Who Was in the Recent Window?
A Rigorous Audit of Online Test-Time Adaptation Privacy

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Abstract

Online test-time adaptation (TTA) updates a deployed model on unlabeled user queries, creating a deployment-time privacy surface absent from frozen inference. We study one exact question: given black-box access to the deployed state f_t at audit time t, can an attacker infer whether a candidate sample appeared in the recent adaptation window $[t-H+1,\ldots,t]$? We instantiate this audit on CIFAR-10-C with two corruptions, three seeds, 5,000-example streams, and three methods: source-only inference, Tent, and T3A. The benchmark combines output-only logistic auditors with unmatched, matched, and nearest-neighbor-matched control pools that remove shortcuts based on class, confidence, stream position, and local feature similarity. Tent delivers large utility gains, improving mean stream accuracy from 0.2794 to 0.7292 on gaussian noise and from 0.6571 to 0.8484 on motion blur. Privacy leakage after strict controls is much weaker: the strongest pooled NN-matched $\mathrm{ROC\text{-}AUC}$ is 0.5589 for T3A on motion blur at $H{=}8$, while Tent stays near chance on gaussian noise and only reaches 0.5441 on motion blur at $H{=}32$. Low-FPR reporting is mostly underpowered under the final matched-pool sizes; at $H{=}32$, the largest pooled TPR@1% FPR is 0.0188. Overall, the benchmark is reproducible and useful, but our confirmatory evidence for strong black-box recent-window leakage is conditional rather than conclusive.

1 Introduction

Online test-time adaptation (TTA) updates deployed model state on unlabeled target inputs to maintain accuracy under distribution shift [**? ? ?**]. That makes deployment-time privacy different from standard training-time membership inference. If the deployed state changes in response to recent queries, then those recent queries may become inferable from the adapted model, even when the original training checkpoint remains fixed.

This paper studies one narrow question:

> Given black-box access to the deployed state f_t at audit time t, can an attacker infer whether a candidate sample $x^\star$ appeared in the recent adaptation window $[t-H+1,\ldots,t]$?

The formulation is deliberately specific. The protected event is indexed by time, so the audited model is also time-indexed: we audit f_t, not a final post-stream model. We do not propose a new TTA algorithm. Instead, we provide a benchmark and protocol for privacy auditing in online TTA, using matched-control design choices motivated by modern low-FPR membership inference practice [**? ?**].

The initial hypothesis behind the project was that at least one standard online TTA method would leak recent-window membership in a way that survives strict matching. The resulting evidence is weaker than that hypothesis. Tent produces large utility gains but little convincing black-box leakage after strict controls. T3A exhibits modest above-chance $\mathrm{ROC\text{-}AUC}$ values, but the signal is inconsistent across horizons and is accompanied by elevated source-only and far-past controls. The paper therefore becomes a rigorous conditional result: under this benchmark, strong black-box recent-window leakage is not yet cleanly established.

Our contributions are:

- We formalize recent-window membership for online TTA as a time-indexed audit problem on the deployed state f_t and instantiate a reproducible black-box benchmark with matched and nearest-neighbor-matched controls.
- We run a complete CIFAR-10-C study over 18 full streams (three methods, two corruptions, three seeds), with 153 audit times per 5,000-example stream and horizons $H \in \{1, 8, 32\}$.

- We show a sharp utility–privacy mismatch: Tent improves mean stream accuracy by 0.4498 on gaussian noise and 0.1913 on motion blur relative to source-only inference, yet strict-control privacy leakage remains weak, with the strongest pooled NN-matched $\mathrm{ROC\text{-}AUC}$ only 0.5589 and the largest pooled TPR@1% FPR only 0.0188.
- We identify a practical evaluation bottleneck: balanced matched pools remove obvious shortcuts, but they leave most low-FPR estimates underpowered at $H{=}1$ and $H{=}8$, so matched and NN-matched $\mathrm{ROC\text{-}AUC}$ become the primary confirmatory endpoints in this benchmark.

Section 2 reviews prior work, Section 3 defines the threat model and audit protocol, Section 4 presents the experimental results, Section 5 discusses limitations, and Section 6 concludes.

2 Related Work

Test-time adaptation. Tent introduced entropy-minimization-based fully test-time adaptation via online updates to batch-normalization affine parameters [**?**]. Subsequent work broadened the design space with continual adaptation [**?**], selective and forgetting-aware updates [**?**], classifier-state adaptation [**?**], nearest-neighbor self-training [**?**], and forward-only adaptation [**?**]. Recent surveys emphasize that test-time adaptation is now a deployment paradigm rather than one algorithm family [**?**]. These papers focus on utility under shift, not on whether recent deployment queries are inferable from the current deployed state.

Membership inference and privacy auditing. Low-FPR evaluation is central in modern privacy auditing. LiRA showed that privacy claims can change substantially when evaluated at strict false-positive rates [**?**], and subsequent work argued that privacy metrics should be analyzed for reliability and disparity rather than treated as single-run point estimates [**?**]. Our work inherits that perspective, but the protected set is not the training set. It is a transient recent deployment window generated by online adaptation.

Adjacent deployment-time privacy. Recent prompt and in-context learning work shows that inference-time context can itself leak private information. This includes locally private in-context learning [**?**], membership inference against in-context examples [**?**], broader privacy analyses of in-context learning [**?**], prompt-membership inference for system prompts [**?**], and differentially private prompt learning [**?**]. These papers are the closest adjacent literature to our setting because they study transient deployment-time state. However, they focus on explicit prompt or context membership. Our question instead concerns implicit state updates induced by online TTA.

Unlike prior work on TTA, our benchmark audits recent deployment queries. Unlike prior work on in-context privacy, our benchmark does not rely on explicit prompt membership. It targets whether online model adaptation leaves a detectable trace of recent users in the deployed state itself.

3 Method

3.1 Threat Model

Let f_{src} be a source model and let an online TTA method process an unlabeled target stream $x_1, x_2, \ldots$, producing deployed states f_t for times $t = 1, 2, \ldots$. For a horizon H, define recent-window membership

$$m_t^H(x^\star) = \begin{cases} 1 & \text{if } x^\star \in \{x_{t-H+1}, \ldots, x_t\}, \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

The audit problem is: from black-box outputs of $f_t(x^\star)$, infer $m_t^H(x^\star)$.

The primary attacker is black-box. It can query f_t on a candidate image and observe class probabilities. It does not know the true label, internal weights, gradients, optimizer state, or adaptation statistics.

3.2 Attack Features and Auditor

For each query, we extract five output-space features from the adapted probabilities:

- maximum predicted probability,
- entropy,
- top-1 minus top-2 margin,
- Gini concentration,
- ℓ_2 norm of log-probabilities.

We train a class-balanced logistic-regression auditor on these five features. Audit times are split contiguously: the earliest 60% are used for training, the next 20% for validation and threshold selection, and the final 20% for held-out testing. This avoids leakage across adjacent deployed states.

3.3 Matched-Control Benchmark Construction

The benchmark is created offline but mirrors the online threat model. At each audit time t, positives are the current recent-window members. For every positive anchor, we construct several negative pools:

- **Unmatched**: random non-members, capped at 32 negatives per positive.
- **Matched**: exact matching on corruption, severity, true class, source predicted class, source-confidence bin, and stream-time bucket, capped at 5 negatives per positive.
- **NN-matched**: the matched pool filtered further by cosine nearest neighbors in frozen source-feature space, keeping the top $k = 5$ neighbors.
- **Far-past**: matched negatives that appeared earlier in the stream but at least $4H$ positions before time t.

True labels are used only for offline benchmark curation, never by the black-box attacker. The confidence bins are three equal-frequency bins computed from source-model confidence, and stream position is coarsened into eight time buckets across the 5,000-example stream.

T3A is selective, so for T3A we additionally match on whether a candidate crosses the adaptation-acceptance threshold in the queried deployed state. This prevents "was admitted for adaptation" from becoming a shortcut.

3.4 TTA Methods Under Audit

Source-only. The source-only baseline performs frozen inference with the same backbone and preprocessing as the adaptive methods. It is the primary null model: in principle, recent-window membership should be difficult to infer because the deployed state does not change.

Tent. Following **?**], Tent updates only batch-normalization affine parameters by minimizing prediction entropy on each incoming batch. In our implementation, non-BN parameters are frozen, batch-normalization running statistics are disabled, and the optimizer is Adam with learning rate 10^{-3}.

T3A. Following **?**], T3A keeps the backbone frozen and adapts classifier supports online. We initialize supports from the source classifier weights, accept new supports only when the maximum class probability is at least 0.15, and keep the 32 lowest-entropy supports per class. This produces a lighter-weight mechanism than gradient-based Tent, but it still changes the deployed decision state over time.

3.5 Evaluation Metrics

We report three classes of metrics.

Table 1: Utility on CIFAR-10-C severity-5 streams. Results are mean ± sample standard deviation over three seeds. Gain is relative to the source-only baseline for the same corruption. Higher is better for all columns except gain can be negative.

Corruption	Method	Mean Stream Acc. ↑	Tail Acc. ↑	Gain vs. Source
Gaussian noise	Source-only	0.2794 ± 0.0026	0.2713 ± 0.0090	0.0000 ± 0.0000
	Tent	$\mathbf{0.7292} \pm \mathbf{0.0080}$	$\mathbf{0.7420} \pm \mathbf{0.0020}$	$\mathbf{0.4498} \pm \mathbf{0.0057}$
	T3A	0.2517 ± 0.0273	0.2727 ± 0.0378	-0.0277 ± 0.0262
Motion blur	Source-only	0.6571 ± 0.0081	0.6627 ± 0.0190	0.0000 ± 0.0000
	Tent	$\mathbf{0.8484} \pm \mathbf{0.0083}$	$\mathbf{0.8467} \pm \mathbf{0.0231}$	$\mathbf{0.1913} \pm \mathbf{0.0038}$
	T3A	0.6587 ± 0.0095	0.6780 ± 0.0092	0.0015 ± 0.0156

Utility. For each run we measure mean stream top-1 accuracy over all $5{,}000$ stream elements, tail accuracy over the final 500 elements, and gain over source-only inference.

Privacy. The main privacy metric is $\mathrm{ROC\text{-}AUC}$. We also report TPR@1% FPR and TPR@0.1% FPR when the validation and test pools each have enough negatives. The benchmark enforces minimum sizes of $10{,}000$ negatives for 1% FPR and $50{,}000$ negatives for 0.1% FPR.

Uncertainty. We use stratified bootstrap confidence intervals over audit times with $1{,}000$ resamples. Bootstrap groups are defined at the audit-time level rather than the candidate-row level to preserve within-state dependence.

4 Experiments

4.1 Setup

The mandatory study uses CIFAR-10-C with two corruptions, `gaussian_noise` and `motion_blur`, both at severity 5. We use a single RobustBench CIFAR-10 corruption model (`Standard`) for all methods and all seeds. Each run replays one stream of length $5{,}000$ sampled without replacement from the $10{,}000$ corrupted test examples. The batch size is 64, the adaptation chunk size is 32, and audit times are every 32 positions from $t = 128$ to $t = 4992$, giving 153 audit times per full stream. We evaluate horizons $H \in \{1, 8, 32\}$ and seeds $\{0, 1, 2\}$.

The environment smoke test reports Python $3.12.7$, PyTorch $2.10.0{+}\mathrm{cu}128$, one NVIDIA RTX A6000 GPU, and four CPU cores. Mean runtime per 5,000-example stream is approximately 88.0–89.2 seconds for source-only, 357.1–369.6 seconds for Tent, and 340.5–352.9 seconds for T3A.

4.2 Main Utility Results

Table 1 shows that the utility behavior is highly method dependent. Tent substantially improves accuracy on both corruptions, while T3A is essentially neutral on motion blur and harmful on gaussian noise. This matters for privacy interpretation: any privacy cost should be judged against the utility gain actually obtained by the adaptation method.

4.3 Main Privacy Results

Table 2 reports pooled matched and NN-matched $\mathrm{ROC\text{-}AUC}$ values. Three observations stand out.

First, strict-control privacy leakage is much weaker than unmatched privacy would suggest. Tent is nearly at chance on gaussian noise for $H = 8$ and $H = 32$ and only modestly above chance on motion blur at $H = 32$.

Second, T3A has the strongest above-chance results, with NN-matched pooled $\mathrm{ROC\text{-}AUC}$ values of 0.5526 on gaussian noise at $H = 8$, 0.5569 on gaussian noise at $H = 32$, 0.5589 on motion blur at $H = 8$, and 0.5513 on motion blur at $H = 32$.

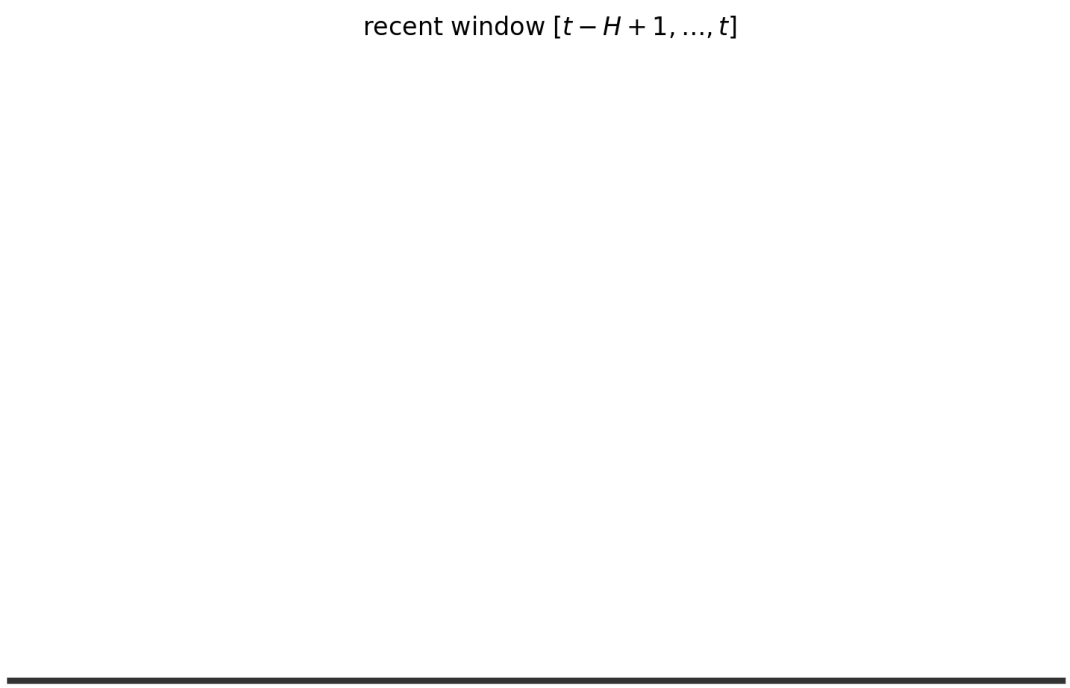

Figure 1: Threat model overview. A fixed source model f_{src} is deployed with online TTA on an unlabeled target stream, producing intermediate deployed states f_t. At audit time t, the attacker queries f_t on a candidate sample and tries to infer whether that sample was present in the recent adaptation window $[t-H+1, \ldots, t]$.

Third, the recency story is not monotone. Our original hypothesis expected the strongest signal at small horizons, but the most stable values often appear at $H = 8$ or $H = 32$ rather than $H = 1$. That weakens the case that the auditor is isolating strictly recent exposure.

The confidence intervals tell the same story. For example, T3A on motion blur at H=8 reaches pooled NN-matched $\mathrm{ROC\text{-}AUC} = 0.5589$ with 95% CI $[0.5391, 0.5780]$, whereas Tent on gaussian noise at H=8 is 0.5073 with 95% CI $[0.4735, 0.5420]$. However, source-only baselines are not perfectly flat either: source-only motion blur reaches NN-matched $\mathrm{ROC\text{-}AUC} = 0.5442$ at H=32. This residual source-only signal is a warning that benchmark construction, not only adaptation, can still leave detectable structure.

4.4 Low-FPR Results

Low-FPR metrics are mostly unavailable under the final matched-pool sizes. The benchmark requires at least $10{,}000$ negatives in both validation and test pools for TPR@1% FPR, which fails for all H=1 and H=8 matched and NN-matched conditions. Only H=32 has enough negatives for pooled TPR@1% FPR, and no condition reaches the $50{,}000$-negative threshold for TPR@0.1% FPR.

Table 3 reports the resulting H=32 values. Even there, the effect sizes are small. The largest pooled NN-matched TPR@1% FPR is 0.0188 for T3A on gaussian noise, and source-only baselines already reach 0.0151 and 0.0168. We therefore do not interpret the low-FPR results as strong evidence of privacy leakage.

Table 2: Pooled privacy results on CIFAR-10-C. Entries are matched and NN-matched pooled ROC-AUC on the held-out test blocks. Best NN-matched value within each corruption is shown in bold. Higher is worse for privacy.

Corruption	Method	M $H{=}1$	NN $H{=}1$	M $H{=}8$	NN $H{=}8$	M $H{=}32$	NN $H{=}32$
Gaussian noise	Source-only	0.5400	0.5367	0.4995	0.5146	0.5166	0.5309
	Tent	**0.5875**	**0.5782**	0.5097	0.5073	0.5061	0.5054
	T3A	0.5424	0.5563	**0.5335**	**0.5526**	**0.5393**	**0.5569**
Motion blur	Source-only	0.4479	0.4641	0.5268	0.5431	0.5303	0.5442
	Tent	**0.5445**	**0.5428**	0.5197	0.5190	0.5280	0.5441
	T3A	0.5137	0.5253	**0.5556**	**0.5589**	**0.5402**	**0.5513**

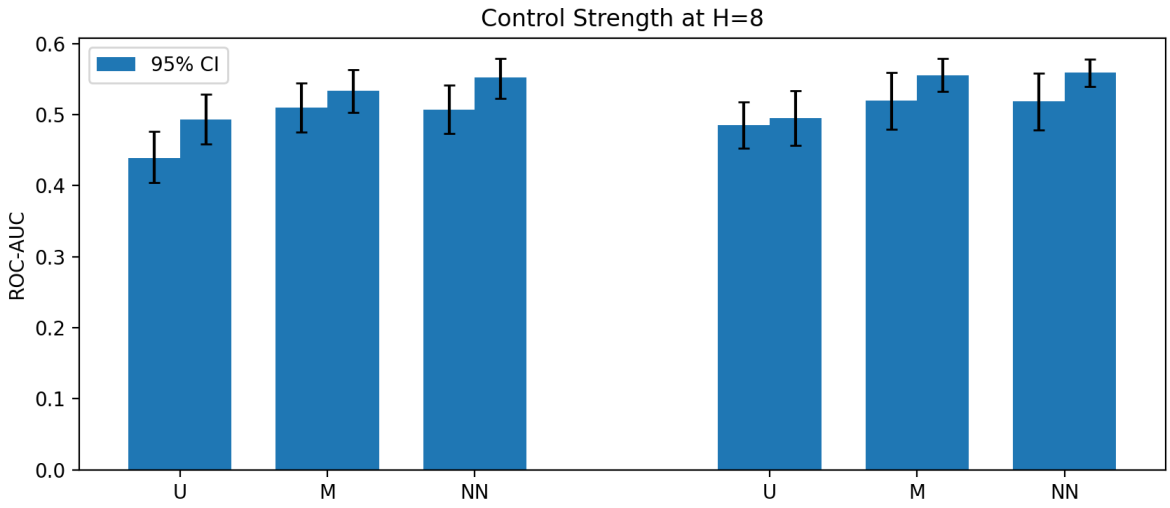

Figure 2: Control-strength comparison at horizon $H{=}8$. Each bar is pooled ROC-AUC for unmatched (U), matched (M), and nearest-neighbor-matched (NN) evaluation. T3A preserves modest signal under stronger controls; Tent is close to chance once matching is applied.

4.5 Ablations and Controls

Figure 4 shows the null controls at $H{=}8$. For Tent, shuffled-membership and far-past controls remain near the matched recent-window values, which supports the interpretation that there is little clean privacy signal after strict control. For T3A, matched recent-window performance is above shuffled nulls, but far-past controls remain competitive, especially on motion blur. That again makes the evidence conditional rather than decisive.

Table 4 summarizes the ablations aggregated over both corruptions and all three seeds. Two trends are clear. First, the multi-feature logistic auditor consistently outperforms the best scalar score, often by more than 0.05–0.10 ROC-AUC points. Second, relaxing individual controls changes the measured privacy only modestly. For T3A at $H{=}8$, the full NN-matched average ROC-AUC is 0.5557, while removing source-prediction matching reduces it to 0.5292 and removing time-bucket matching reduces it to 0.5245. For Tent, all variants remain near chance.

5 Discussion and Limitations

The strongest empirical finding in this study is not a privacy break. It is the gap between utility and privacy evidence. Tent produces clear utility benefits, but once we enforce matched and NN-matched controls, its privacy signal is usually near chance. T3A shows more consistent above-chance ROC-AUC, but the effect is modest and does not clearly track recency. That matters because the motivating claim was specifically about *recent-window* membership.

Several limitations follow directly from the current workspace.

Table 3: Pooled TPR@1% FPR at horizon H=32. Entries are reported only when both validation and test pools contain at least 10,000 negatives. Higher is worse for privacy.

Corruption	Method	Matched	NN-matched
Gaussian noise	Source-only	0.0108	0.0151
	Tent	0.0097	0.0114
	T3A	0.0141	**0.0188**
Motion blur	Source-only	0.0134	0.0168
	Tent	0.0134	**0.0181**
	T3A	—	0.0128

Table 4: Ablation summary aggregated over both corruptions and all three seeds. "Full" is the black-box NN-matched ROC-AUC. The remaining columns remove one control from benchmark construction or replace the logistic auditor with the best scalar score chosen on seed 0. Higher values indicate worse privacy.

Method	H	Full	−TrueCls	−Time	−SrcPred	−Conf	−NN	Best Scalar
Tent	8	0.5132	0.5322	0.5152	0.5121	0.5167	0.5190	0.4776
Tent	32	0.5247	0.5129	0.5217	0.5262	0.5325	0.5200	0.4695
T3A	8	**0.5557**	0.5355	0.5245	0.5292	0.5469	0.5452	0.4927
T3A	32	**0.5541**	0.5370	0.5256	0.5328	0.5496	0.5411	0.4888

Low-FPR underpowering. The final balanced matched pools are too small for most low-FPR estimates. This is not a reporting choice; it is enforced by the minimum-negative thresholds built into the analysis pipeline. As a result, matched and NN-matched $\mathrm{ROC\text{-}AUC}$ become the primary confirmatory endpoints, and the strongest privacy claims that can be made are limited.

Residual baseline structure. Source-only and far-past controls are not perfectly flat. For example, source-only motion blur reaches NN-matched $\mathrm{ROC\text{-}AUC} = 0.5442$ at H=32. This suggests that stream position, image hardness, or source-feature locality are not fully neutralized by the current benchmark design.

Scope. The study is intentionally narrow: CIFAR-10-C only, two corruptions, one severity, one backbone, and three methods. We do not claim transfer to other datasets, larger models, or modalities. We also do not evaluate stronger gray-box or white-box attackers.

Benchmark rather than proof of non-leakage. Near-chance black-box results do not prove that online TTA is private. They only show that, under this attack interface and benchmark construction, strong recent-window leakage was not cleanly demonstrated.

The broader lesson is methodological. Deployment-time privacy audits need the same discipline as modern training-time membership inference: exact threat models, strict matching, null controls, and enough negatives for low-FPR evaluation. Without those ingredients, it is too easy to confuse incidental structure with privacy leakage.

6 Conclusion

We presented a benchmark for auditing whether a candidate sample appeared in the recent adaptation window of an online TTA model state f_t. The benchmark fixes the threat model, enforces matched and nearest-neighbor-matched controls, and evaluates black-box auditors on CIFAR-10-C streams.

The resulting evidence is mixed and mostly conservative. Tent improves mean stream accuracy from 0.2794 to 0.7292 on gaussian noise and from 0.6571 to 0.8484 on motion blur, but its strict-control privacy signal is weak. T3A reaches the highest pooled NN-matched $\mathrm{ROC\text{-}AUC}$ values, up to 0.5589, yet low-FPR results remain small and

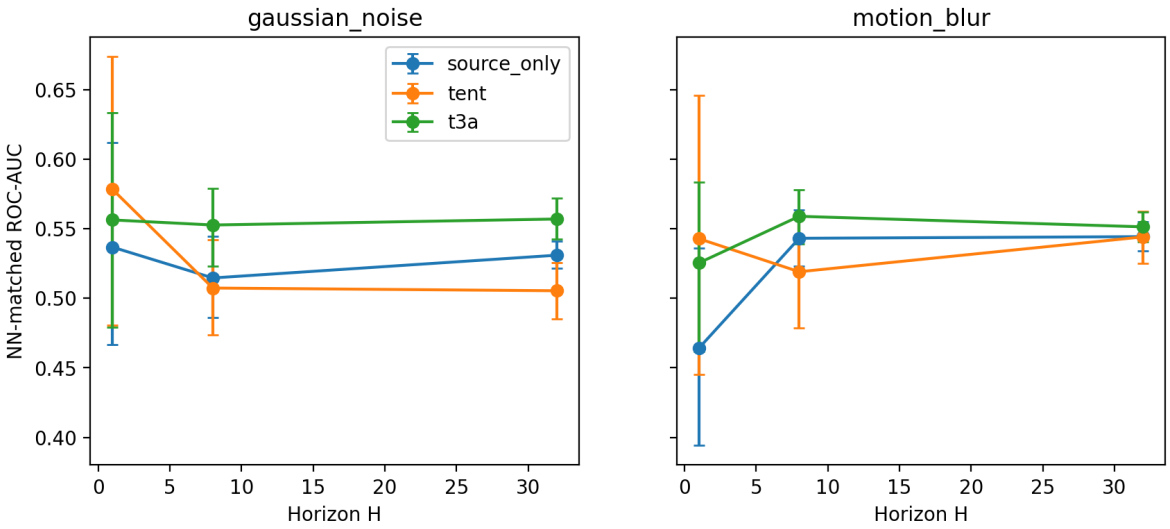

Figure 3: NN-matched ROC-AUC versus horizon. The expected monotone recency decay is not observed cleanly. Tent is near chance on gaussian noise across all horizons, whereas T3A stays modestly above chance across $H \in \{8, 32\}$ on both corruptions.

source-only controls are not flat. We therefore view the current paper as a rigorous conditional audit result rather than a strong positive privacy attack.

Future work should scale the benchmark to larger pools, stronger attackers, and more realistic deployment settings so that low-FPR evaluation becomes routine rather than optional. That is the main path toward deciding whether recent deployment queries in online adaptation are measurably exposed in practice.

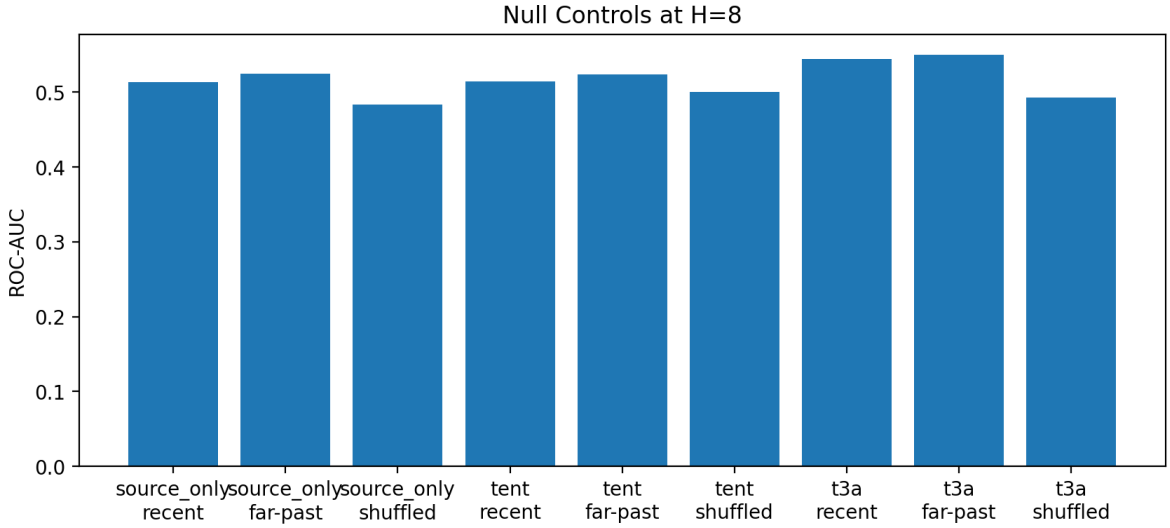

Figure 4: Null controls at $H{=}8$. For each method, we compare matched recent-window auditing against a far-past control and a shuffled-membership null. The gap between recent and null conditions is small for Tent and only moderate for T3A, especially once far-past controls are included.